

SYSTEM ADMINISTRATION

ENTERPRISE SECURITY

RELEASE V2.4

OCPP Charge Point Management System

Enterprise Administration, Multi-Tenancy, Granular RBAC, PKI Certificates, EPEX Tariffs, Auto-Heal & Live Protocol Diagnostics

TARGET AUDIENCE

System Administrators, Security Officers & Technical CPOs

PUBLICATION DATE

August 2026

SECURITY COMPLIANCE

OCPP 1.6 Security Profile 3 (mTLS) & ISO 15118

PUBLISHER

webdotpulse / Mobility Pulse Network

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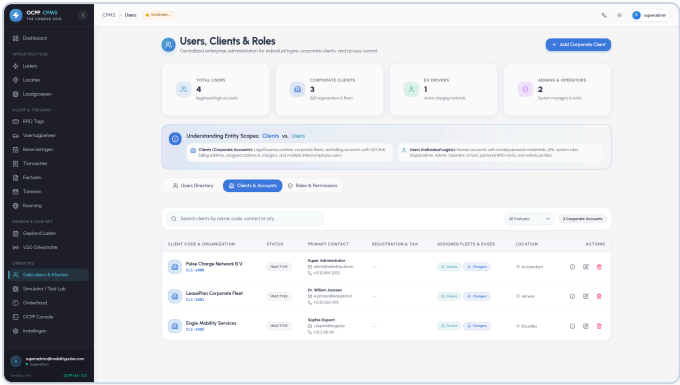
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1. Multi-Tenant Architecture & Corporate Hierarchy

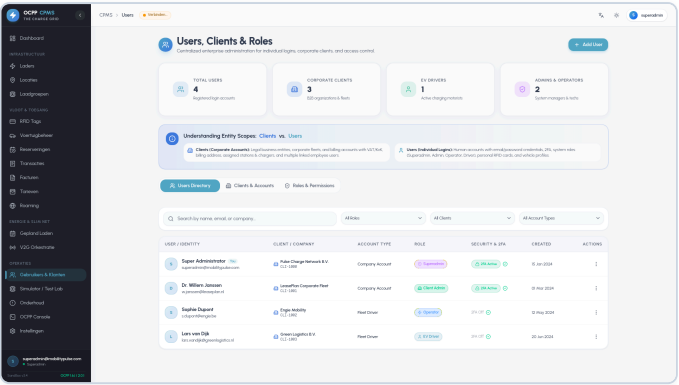
The CPMS architecture provides strict logical isolation between independent corporate clients, billing entities, and operational domains.

MULTI-TENANCY ISOLATION RULE

All database queries for non-superadmin users are strictly scoped by `companyId` or `owner_id`. Corporate client administrators can never view or modify chargers assigned to other organizations.



Corporate B2B Clients Directory



User Accounts & Roles Directory

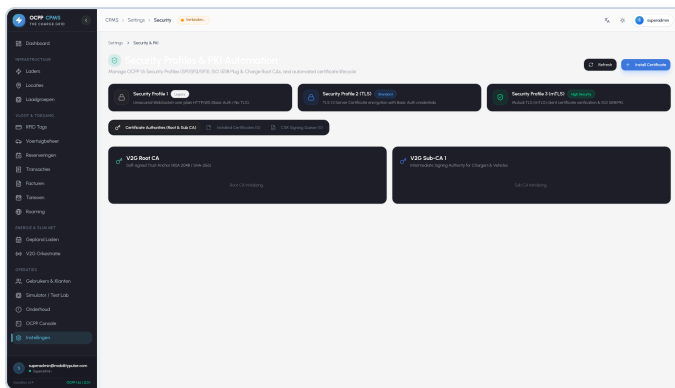
Module	SuperAdmin	Platform Admin	Operator/Tech	Client Admin	User/Driver
Infrastructure & Stations	Full CRUD	Full CRUD	View & Diagnostics	Assigned Only	Map View
Remote Controls	Full Access	Full Access	Full Access	Restricted	Own Socket
Tariffs & Dynamic EPEX	Full CRUD	Full CRUD	View Only	View Assigned	View Rates
Invoices & SEPA Direct Debit	Full CRUD	Full CRUD	No Access	Own Invoices	Own Receipts
Roaming Hubs (OCPI/OICP)	Full CRUD	Manage	View Only	No Access	No Access
PKI Security & Audit Trail	Full CRUD	View Audit	No Access	No Access	No Access

Interactive Roles & Permissions Matrix Console

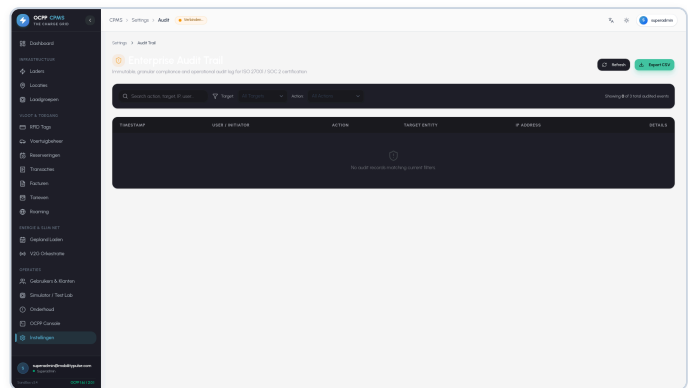
3. Security Profiles & PKI / TLS Certificates

The CPMS enforces standards compliant with **OCPP 1.6 Security Profile 3** and **ISO 15118 PKI**:

- **Profile 1:** Unencrypted HTTP / WS transport (restricted to private VPNs).
- **Profile 2:** TLS / WSS transport with HTTP Basic Authentication.
- **Profile 3:** Mutual TLS (mTLS) with bidirectional X.509 client and server certificates.
- **Certificate Authority Trust Store:** Upload and inspect Root CAs, Sub-CAs, and revocation lists.



PKI Security Profiles & X.509 Management



Enterprise Audit Trail & Compliance Log

4. Enterprise Audit Trail & Compliance

Every administrative action, remote control RPC invocation, and tariff modification is recorded immutably in the audit trail with UTC timestamp, actor IP, user-agent, target resource, and outcome status.

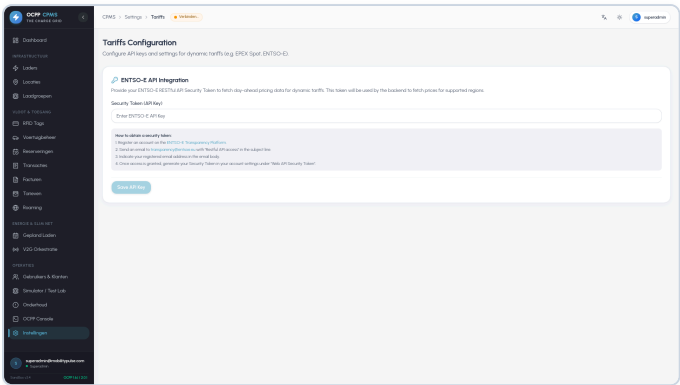
5. Dynamic EPEX Spot Pricing & Market Feeds

The **Dynamic Tariffs Engine** (`/settings/tariffs`) connects with wholesale Day-Ahead electricity markets to compute real-time hourly consumer tariffs:

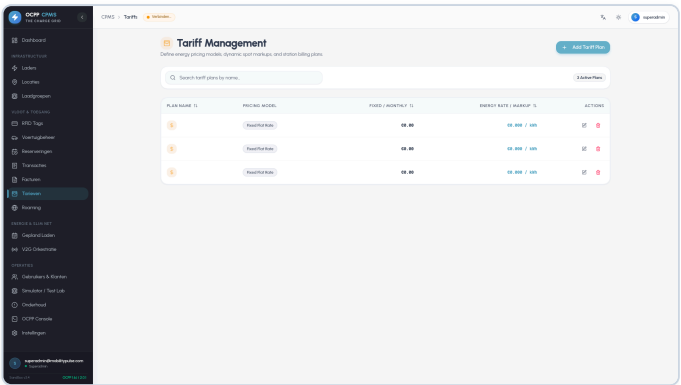
SPOT PRICING FORMULA

$$\text{Tariff (€/kWh)} = (\text{Spot Price} \times \text{Multiplier}) + \text{CPD Markup (€/kWh)} + \text{Grid Fee} + \text{VAT (21\%)}$$

- **Supported Market Feeds:** EnergyZero API, ENTSO-E Transparency Platform, and Energy-Charts API.
- **Automated Cron:** Daily price ingestion fires at 13:15 CET for the subsequent 24-hour delivery period.



Dynamic EPEX Spot Tariffs Settings



Tariffs Overview Matrix

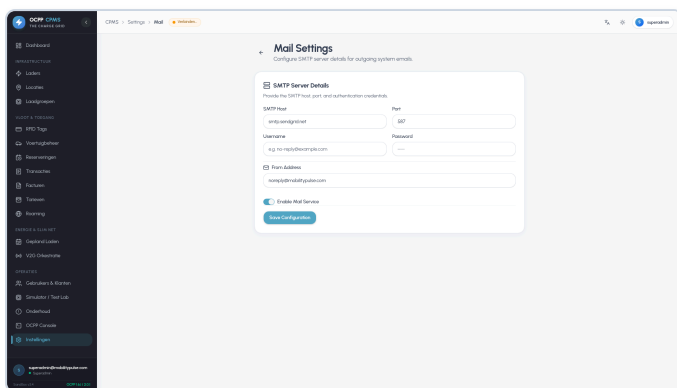
6. SMTP Mail Server & HTML Template Engine

6.1 Outgoing SMTP Configuration (/settings/mail)

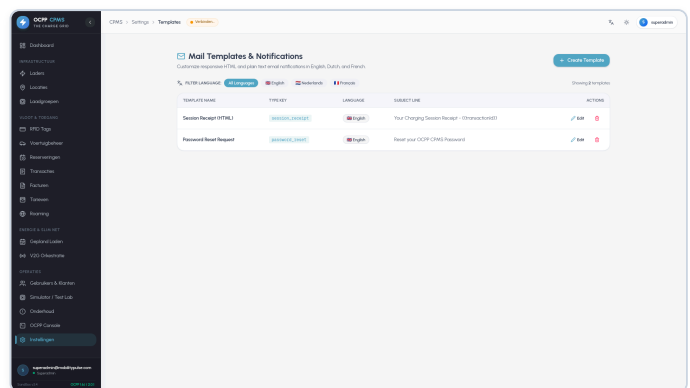
Configure transactional email delivery via Postmark, SendGrid, Amazon SES, or custom SMTP relays with TLS encryption and testing tools.

6.2 HTML Mail Template Editor (/settings/templates)

Customize email templates with visual previews and dynamic variables: `{{user_name}}` , `{{invoice_number}}` , `{{total_amount}}` , `{{kwh_delivered}}` .



SMTP Outgoing Mail Server Configuration

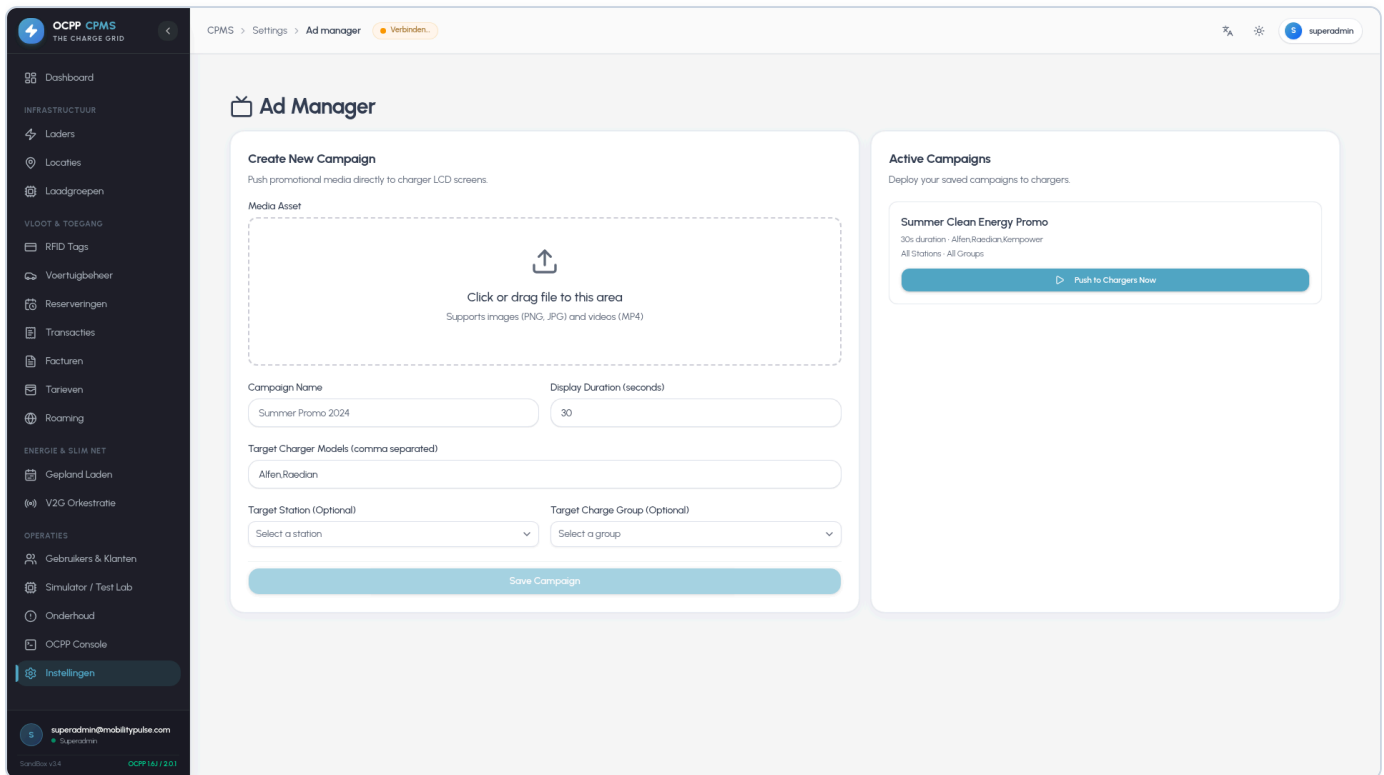


HTML Mail Template Visual Editor

7. Screen Advertising Manager & Media Campaigns

Distribute digital advertisement media campaigns to chargers equipped with color LCD screens:

- Supports MP4 (H.264 video), PNG, JPG, and dynamic HTML5 promo banners.
- Target campaigns by specific stations, charge groups, or geographic cities.
- Distribute assets via customized OCPP `DataTransfer` vendor RPCs.

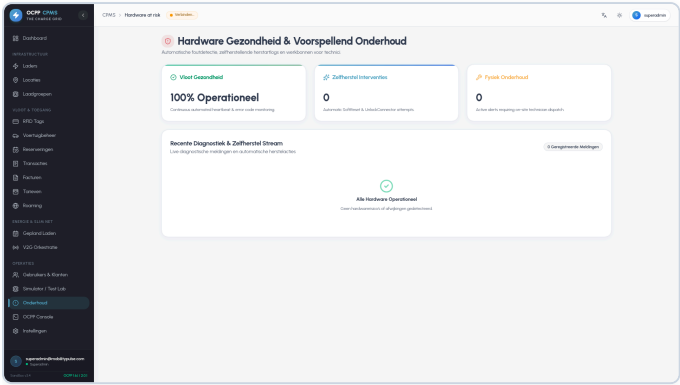


Screen Advertising Campaign Scheduler

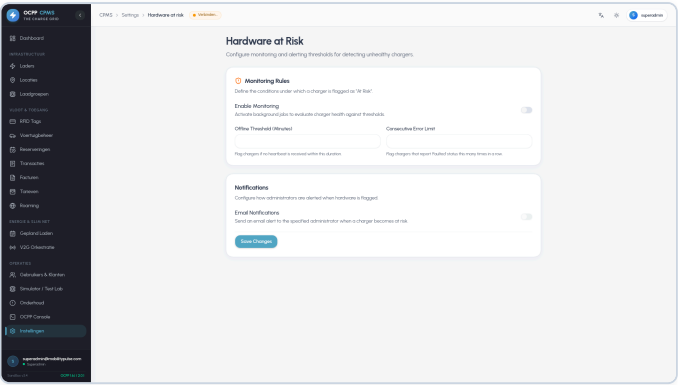
8. Hardware-at-Risk Engine & Auto-Heal Rules

The **Hardware-at-Risk Subsystem** (`/hardware-at-risk` & `/settings/hardware-at-risk`) continuously evaluates telemetry to resolve hardware failures automatically:

AUTO-HEAL PROTOCOL
If a charger misses heartbeats for >180 seconds or experiences a suspended EVSE lock, the engine automatically issues a remote soft reset before escalating an incident ticket to field technicians.



Hardware-at-Risk Live Fleet Monitor



Auto-Heal Heuristic Rules Configuration

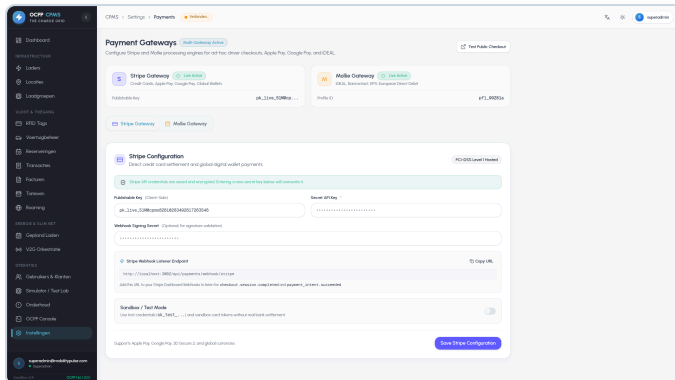
9. Payment Gateways & Roaming Connections

9.1 Stripe & Mollie Gateway Setup (/settings/payments)

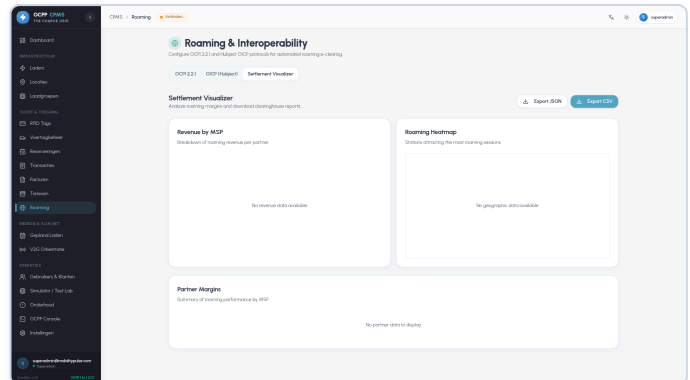
Manage API credentials, test sandbox modes, and webhook signing secrets for instant public ad-hoc payment processing.

9.2 Roaming Hubs (OCPI 2.2.1 & Hubject OICP)

Manage bilateral roaming credentials, exchange tokens, and track wholesale clearinghouse settlements.



Mollie & Stripe Payment Gateway Credentials



Roaming Settlement & Clearinghouse Visualizer

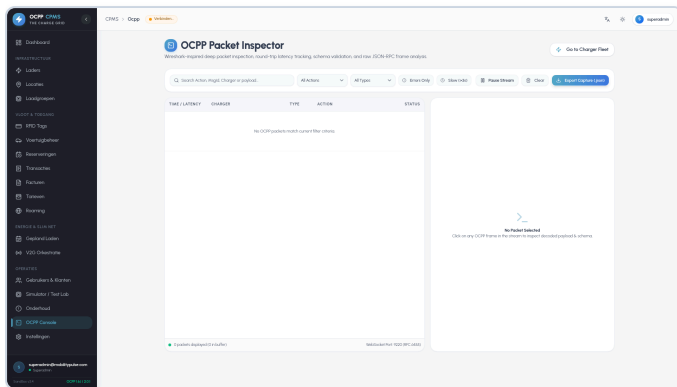
10. Live OCPP Packet Inspector & Quirk Profiles

10.1 Live Packet Inspector (/ocpp)

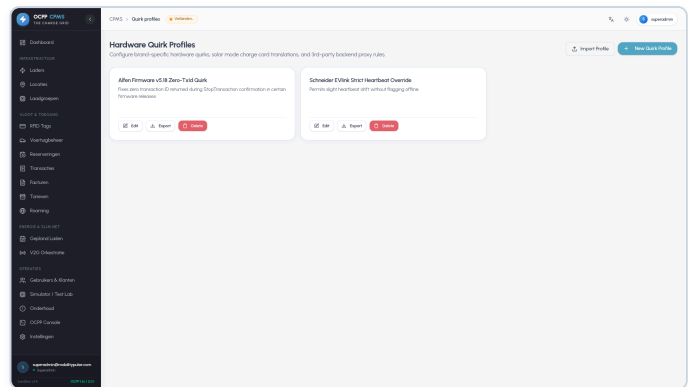
Stream real-time unbuffered WebSocket JSON-RPC frames (`CALL` , `CALLRESULT` , `CALLERROR`) with full syntax parsing and manual RPC dispatcher.

10.2 Hardware Quirk Profiles (/quirk-profiles)

Non-intrusively repair vendor-specific OCPP firmware non-compliance (missing power derivation, integer multipliers, UID endianness).



Live OCPP Packet Inspector & RPC Console



Hardware Quirk Profiles & Overrides